Pattern Program 1: Right-Angled Triangle of Stars

🔹 Problem Statement:

Write a Java program to print a right-angled triangle of * stars for a given number of rows.

🔹 Sample Output (for 5 rows):

```
*
**
***
****
*****
```

🔹 Java Program:

```
import java.util.Scanner;

public class RightAngledTriangle {
    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.print("Enter number of rows: ");
        int rows = sc.nextInt();

        for(int i = 1; i <= rows; i++) {
```

```
            for(int j = 1; j <= i; j++) {
                System.out.print("*");
            }

            System.out.println();
        }

        sc.close();
    }
}
```

Pattern Program 2: Number Pyramid

🔹 Problem Statement:

Write a Java program to print a number pyramid pattern based on the number of rows.

🔹 Sample Output (for 5 rows):

```
    1
   1 2
  1 2 3
 1 2 3 4
1 2 3 4 5
```

🔹 Java Program:

```
import java.util.Scanner;
```

```
public class NumberPyramid {
    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.print("Enter number of rows: ");
        int rows = sc.nextInt();

        for(int i = 1; i <= rows; i++) {

            for(int space = 1; space <= rows - i; space++) {
                System.out.print(" ");
            }

            for(int j = 1; j <= i; j++) {
                System.out.print(j + " ");
            }

            System.out.println();
        }

        sc.close();
    }
}
```

1. Hollow Square Pattern

🔹 Problem Statement:

Write a Java program to print a hollow square pattern using stars.

- Sample Output:

```
*****
*   *
*   *
*   *
*****
```

- Java Program:

```
public class HollowSquare {
    public static void main(String[] args) {

        for(int i = 1; i <= 5; i++) {

            for(int j = 1; j <= 5; j++) {

                if(i == 1 || i == 5 || j == 1 || j == 5)
                    System.out.print("*");
                else
                    System.out.print(" ");
            }

            System.out.println();
        }
    }
}
```

2. Number Right-Angled Triangle

- Problem Statement:

Write a Java program to print a right-angled triangle using numbers.

- Sample Output:

```
1
1 2
1 2 3
1 2 3 4
1 2 3 4 5
```

- Java Program:

```
public class NumberRightTriangle {
    public static void main(String[] args) {

        for(int i = 1; i <= 5; i++) {

            for(int j = 1; j <= i; j++) {
                System.out.print(j + " ");
            }

            System.out.println();
        }
    }
```

```
}
```

3. Alphabet Triangle Pattern

🔹 Problem Statement:

Write a Java program to print an alphabet triangle pattern.

🔹 Sample Output:

```
A
A B
A B C
A B C D
A B C D E
```

🔹 Java Program:

```
public class AlphabetTriangle {
    public static void main(String[] args) {

        for(int i = 1; i <= 5; i++) {

            char ch = 'A';

            for(int j = 1; j <= i; j++) {
                System.out.print(ch + " ");
                ch++;
            }
```

```
            System.out.println();
        }
    }
}
```

4. Diamond Star Pattern

◆ Problem Statement:

Write a Java program to print a diamond pattern using stars.

◆ Sample Output:

```
  *
 ***
*****
 ***
  *
```

◆ Java Program:

```
public class DiamondPattern {
    public static void main(String[] args) {

        int n = 3;

        for(int i = 1; i <= n; i++) {
```

```
            for(int space = 1; space <= n - i; space++)
                System.out.print(" ");

            for(int j = 1; j <= 2 * i - 1; j++)
                System.out.print("*");

            System.out.println();
        }


        for(int i = n - 1; i >= 1; i--) {

            for(int space = 1; space <= n - i; space++)
                System.out.print(" ");

            for(int j = 1; j <= 2 * i - 1; j++)
                System.out.print("*");

            System.out.println();
        }
    }
}
```

5. Floyd’s Triangle

- Problem Statement:

Write a Java program to print Floyd’s triangle.

- Sample Output:

1

2 3

4 5 6

7 8 9 10

11 12 13 14 15

🔹 Java Program:

```
public class FloydTriangle {
    public static void main(String[] args) {

        int num = 1;

        for(int i = 1; i <= 5; i++) {

            for(int j = 1; j <= i; j++) {
                System.out.print(num + " ");
                num++;
            }

            System.out.println();
        }
    }
}
```

Pattern Program 7: Right-Aligned Star Triangle

🔹 Problem Statement:

Write a Java program to print a right-aligned triangle pattern using stars.

🔹 Sample Output (for 5 rows):

```
    *
   **
  ***
 ****
*****
```

🔹 Java Program:

```
public class RightAlignedStarTriangle {
    public static void main(String[] args) {

        for(int i = 1; i <= 5; i++) {

            for(int space = 1; space <= 5 - i; space++) {
                System.out.print(" ");
            }

            for(int j = 1; j <= i; j++) {
                System.out.print("*");
            }

            System.out.println();
        }
```

```
    }
}
```

Pattern Program 8: Inverted Number Triangle

- Problem Statement:

Write a Java program to print an inverted number triangle pattern.

- Sample Output:

```
1 2 3 4 5
1 2 3 4
1 2 3
1 2
1
```

- Java Program:

```
public class InvertedNumberTriangle {
    public static void main(String[] args) {

        for(int i = 5; i >= 1; i--) {

            for(int j = 1; j <= i; j++) {
                System.out.print(j + " ");
            }

            System.out.println();
```

```
        }
    }
}
```

Pattern Program 9: Binary Number Triangle

🔹 Problem Statement:

Write a Java program to print a binary number triangle pattern.

🔹 Sample Output:

```
1
0 1
1 0 1
0 1 0 1
1 0 1 0 1
```

🔹 Java Program:

```
public class BinaryNumberTriangle {
    public static void main(String[] args) {

        for(int i = 1; i <= 5; i++) {

            for(int j = 1; j <= i; j++) {

                System.out.print((i + j) % 2 + " ");
            }
```

```
            System.out.println();
        }
    }
}
```

Pattern Program 10: Half Diamond Star Pattern

- Problem Statement:

Write a Java program to print a half diamond pattern using stars.

- Sample Output:

```
*
**
***
**
*
```

- Java Program:

```
public class HalfDiamondStar {
    public static void main(String[] args) {

        for(int i = 1; i <= 3; i++) {

            for(int j = 1; j <= i; j++) {
                System.out.print("*");
```

```
            }

            System.out.println();
        }


        for(int i = 2; i >= 1; i--) {

            for(int j = 1; j <= i; j++) {
                System.out.print("*");
            }

            System.out.println();
        }
    }
}
```

Pattern Program 11: Hollow Right-Angled Triangle

- Problem Statement:

Write a Java program to print a hollow right-angled triangle using stars.

- Sample Output:

```
*
**
* *
*  *
*****
```

◆ Java Program:

```
public class HollowRightTriangle {
    public static void main(String[] args) {

        for(int i = 1; i <= 5; i++) {

            for(int j = 1; j <= i; j++) {

                if(j == 1 || j == i || i == 5)
                    System.out.print("*");
                else
                    System.out.print(" ");
            }

            System.out.println();
        }
    }
}
```

Pattern Program 12: Reverse Pyramid of Stars

◆ Problem Statement:

Write a Java program to print a reverse pyramid pattern using stars.

◆ Sample Output:

```
*********
 *******
  *****
   ***
    *
```

- Java Program:

```java
public class ReversePyramidStars {
    public static void main(String[] args) {

        int rows = 5;

        for(int i = rows; i >= 1; i--) {

            for(int space = 0; space < rows - i; space++) {
                System.out.print(" ");
            }

            for(int j = 1; j <= 2 * i - 1; j++) {
                System.out.print("*");
            }

            System.out.println();
        }
    }
}
```

Pattern Program 13: Continuous Characters Triangle

🔹 Problem Statement:

Write a Java program to print a continuous characters triangle pattern.

🔹 Sample Output:

```
A
B C
D E F
G H I J
K L M N O
```

🔹 Java Program:

```
public class ContinuousCharactersTriangle {
    public static void main(String[] args) {

        char ch = 'A';

        for(int i = 1; i <= 5; i++) {

            for(int j = 1; j <= i; j++) {
                System.out.print(ch + " ");
                ch++;
            }
```

```
            System.out.println();
        }
    }
}
```

Pattern Program 14: Right-Angled Triangle with Even Numbers

🔹 Problem Statement:

Write a Java program to print a right-angled triangle pattern using even numbers.

🔹 Sample Output:

```
2
2 4
2 4 6
2 4 6 8
2 4 6 8 10
```

🔹 Java Program:

```
public class EvenNumberTriangle {
    public static void main(String[] args) {

        for(int i = 1; i <= 5; i++) {

            for(int j = 1; j <= i; j++) {
                System.out.print((2 * j) + " ");
            }
```

```
            System.out.println();
        }
    }
}
```

Pattern Program 15: Triangle of Odd Numbers

- Problem Statement:

Write a Java program to print a triangle pattern using consecutive odd numbers.

- Sample Output:

```
1
3 5
7 9 11
13 15 17 19
21 23 25 27 29
```

- Java Program:

```
public class OddNumberTriangle {
    public static void main(String[] args) {

        int num = 1;

        for(int i = 1; i <= 5; i++) {
```

```
            for(int j = 1; j <= i; j++) {

                System.out.print(num + " ");
                num += 2;
            }

            System.out.println();
        }
    }
}
```

Pattern Program 16: Hollow Pyramid Pattern

- Problem Statement:

Write a Java program to print a hollow pyramid pattern using stars.

- Sample Output:

```
    *
   * *
  *   *
 *     *
*********
```

- Java Program:

```
public class HollowPyramid {
    public static void main(String[] args) {
```

```
        int rows = 5;

        for(int i = 1; i <= rows; i++) {

            for(int space = 1; space <= rows - i; space++) {
                System.out.print(" ");
            }

            for(int j = 1; j <= 2 * i - 1; j++) {

                if(j == 1 || j == 2 * i - 1 || i == rows)
                    System.out.print("*");
                else
                    System.out.print(" ");
            }

            System.out.println();
        }
    }
}
```

Pattern Program 17: Floyd’s Triangle (Using User Input)

- Problem Statement:

Write a Java program to print Floyd’s triangle for a given number of rows.

- Sample Output (for 5 rows):

1

2 3

4 5 6

7 8 9 10

11 12 13 14 15

- Java Program:

```
import java.util.Scanner;

public class FloydTriangleInput {
    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.print("Enter number of rows: ");
        int rows = sc.nextInt();

        int num = 1;

        for(int i = 1; i <= rows; i++) {

            for(int j = 1; j <= i; j++) {
                System.out.print(num + " ");
                num++;
            }

            System.out.println();
```

```
        }

        sc.close();
    }
}
```